

Modeling, mapping & targeting **neuropsychiatric disease** in patient-derived brain organoids

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Neuropsychiatric disorders (NPDs) are heterogeneous, polygenic, and still treated with one-size-fits-all drugs. I will turn living patient brain organoids into a **mechanistic and therapeutic map** of NPDs.

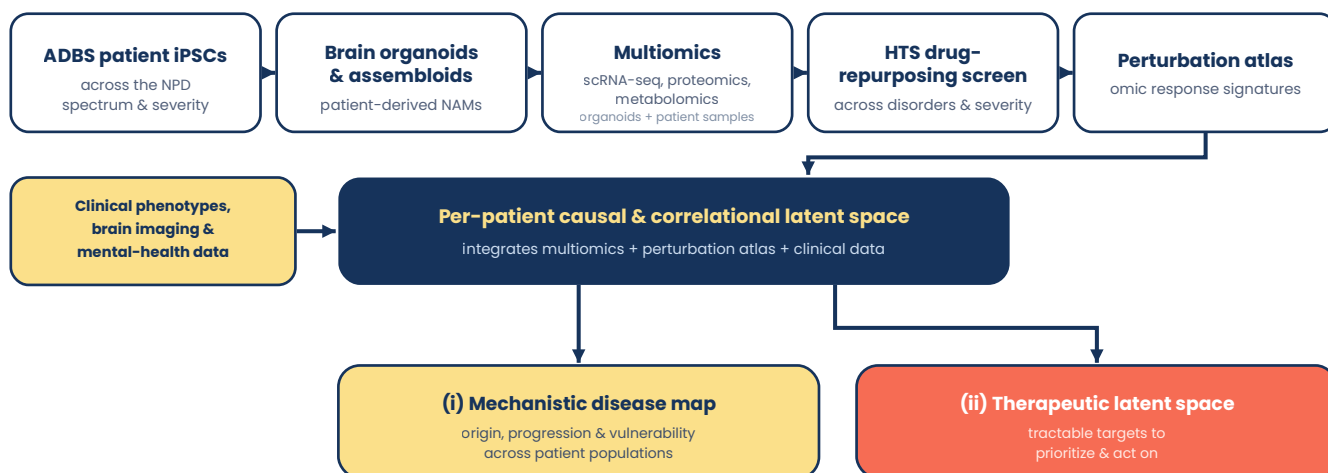
Approach

From **ADBS** patient iPSCs, I generate brain-region organoids and assembloids across the NPD spectrum and severity. **Multimomics** from organoids and patient samples define per-patient molecular states, embedded into **causal & correlational latent spaces**. A **high-throughput drug-repurposing screen** then builds a **perturbation atlas**, integrated with patient phenotypes, brain imaging, and mental-health data.

Aims

- **i. Map disease mechanism.** Build an informed map of NPD **origin, progression & vulnerability** across patient populations.
- **ii. Map the therapeutic landscape.** Construct a **latent-space representation** of tractable targets to identify and act on.

Platform at a glance



From ADBS patient iPSCs to a per-patient latent space that yields a mechanistic disease map (i) and an actionable therapeutic landscape (ii).

★ Feasibility

Established human iPSC **organoid & assembloid** expertise, **high-throughput screening**, and **AI/ML-assisted** multimomics & imaging analysis.

★ Impact

A reusable **patient-to-target engine** for precision psychiatry, centering Indian patient cohorts.

★ Institutional partnership

Built with **ICMR, NIMHANS, CBM**, and the **CALM-Brain database**, under ICMR-mandated ethics, consent & biosample governance.